

KAVISH SHANILKA

Electrical Engineering Undergraduate
University of Moratuwa, Sri Lanka

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ABOUT ME

Electrical engineering undergraduate at the University of Moratuwa passionate about research, intelligent systems, embedded hardware, and power electronics. Thrives in collaborative environments solving complex engineering challenges through innovative instrumentation and control systems.

Areas of Interest: Power Electronics, Embedded Systems, Computer Vision, Microgrids, Autonomous Systems, Digital Signal Processing, IoT, Robotics, Renewable Energy

EDUCATION

University of Moratuwa

B.Sc. Hons in Electrical Engineering (Undergraduate)

Sri Lanka
Feb. 2024 – Present

- Cumulative GPA: 3.85 / 4.00

Bandaranayake Central College, Veyangoda

G.C.E. Advanced Level Examination (Physical Science Stream)

Sri Lanka
2011 – 2019

- Z-Score: 2.6192 (3 As – Combined Mathematics, Physics, Chemistry)
- Ranked 131st in the country (District Rank: 12th)

AWARDS AND SCHOLARSHIPS

Semi-Finalist – Spark Challenge | *Spark Challenge Organizing Committee*

2025 – 2026

- Selected for semi-finals for designing **GridGuard**, an edge-coordinated smart energy management system.

University Category Competitor – SLIOT Challenge 2026 | *CSE Dept. (UoM), SLT-Mobitel, IESL* April 2026

- Competed as part of Team Neura in the national IoT innovation challenge with an automated hardware prototype.

Rotaractor Award | *Rotaract Club*

2025

- Honored in recognition of dedicated service, leadership, and active contributions across key community projects.

PROJECTS

- **Smart Automatic Power Factor Corrector (APFC) Panel** 
 - Engineered automated Power Factor Corrector panel featuring CT/VT sensing circuits and dynamic capacitor bank switching.
 - Built custom LabVIEW dashboards with NI DAQ card and ESP32 OLED Telegram alerts for real-time status and safety monitoring.
- **GridGuard – Autonomous Edge-Coordinated Energy Management System**
 - Designed distributed IoT smart grid manager with Raspberry Pi central hub and ESP32 edge nodes for Spark Challenge.
 - Implemented non-invasive main DB power auditing and peak load-shifting algorithms, reducing electricity costs by > 30%.
- **Smart Irrigation System Controller (MATLAB / Simulink)** 
 - Modeled closed-loop plant dynamics and hysteresis controller logic for automated multi-zone water storage and routing.

- Integrated multi-zone moisture routing algorithms and fault detection logic to optimize water and energy consumption.
- **Advanced Light Intensity Indicator (ALII) – Discrete Digital Hardware** [🔗](#)
 - Designed discrete hardware signal processor without microcontrollers to monitor and average light intensity in real-time.
 - Implemented analog filtering for 50–100 Hz harmonics and built flip-flop logic for time-windowed averaging (300–900 s).

EXPERIENCE

Design Team Lead

March 2026 – April 2026

Prakampana '26 – Department of Electrical Engineering, UoM

Moratuwa, Sri Lanka

- Directed visual design branding, social media campaign artwork, and promotional assets for the annual departmental flagship event.

Public Relations Pillar Member

2025 – Present

Institution of Engineers Sri Lanka (IESL) Young Members Section

Moratuwa, Sri Lanka

- Executed digital outreach campaigns and coordinated event communication strategies for the student chapter.

Project Chair – Project HOPE

2025

Leo Club

Sri Lanka

- Spearheaded community empowerment initiatives targeting sustainable income generation pathways for underprivileged groups.

Project Chair – CAST 4

2025

Rotaract Club

Sri Lanka

- Organized the onboarding and networking gathering for new members, leading logistics, team-building sessions, and entertainment.

SKILLS SUMMARY

- **Languages:** English (professional proficiency), Sinhala (native proficiency)
- **Technical Domains:** Power Electronics, Power System Optimization, Embedded Systems, IoT, Control Engineering, DSP
- **Hardware & Embedded:** ESP32, Raspberry Pi, NI DAQ, Arduino, Sensor Interfacing (CT/VT, PZEM), Relay Logic
- **Software & Simulation:** MATLAB, Simulink, LabVIEW, LTSpice, Proteus, SolidWorks, Git
- **Programming Languages:** C, C++, Python, Java, MATLAB Scripting
- **Soft Skills:** Technical Leadership, Team Collaboration, Project Management, Analytical Problem Solving

REFERENCES

Prof. Logeeshan Velmanickam

Senior Lecturer, Department of Electrical Engineering
University of Moratuwa, Sri Lanka

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